

IB AAHL MATHEMATICS

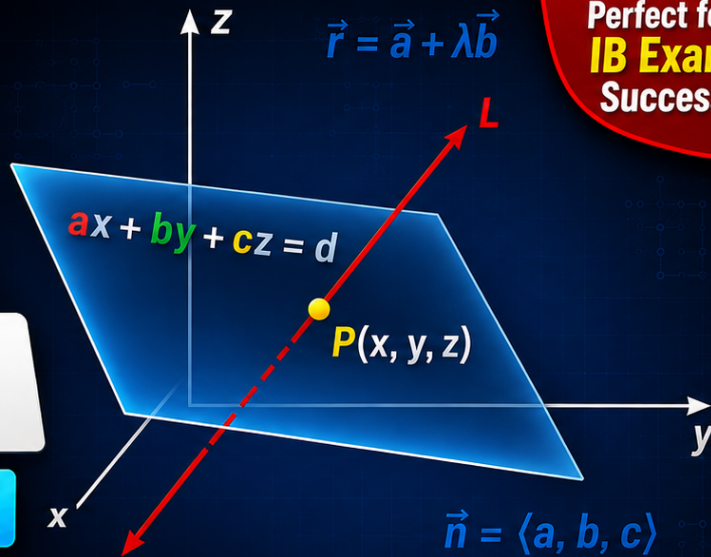
$$\vec{n} \cdot \vec{r} = d$$

HOW TO FIND THE POINT OF INTERSECTION

OF A PLANE AND A LINE

3D

IN 4 EASY STEPS



Perfect for
IB Exam
Success!



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Concepts
Made Easy



Exam
Focused



Step-by-Step
Explanation



Point of intersection of a line and a plane in IB DP Mathematics:

AHL 3.18—Intersections of lines & planes

Date November 2020

Marks available 4

Reference code 20N.2.AHL.TZ0.H_10

Level Additional Higher Level

Paper Paper 2

Time zone Time zone 0

Command term Find

Question number H_10

Adapted from N/A

Question

The plane Π_1 has equation $3x - y + z = -13$ and the line L has vector equation

$$\mathbf{r} = \begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix} + \lambda \begin{pmatrix} -3 \\ -1 \\ 4 \end{pmatrix}, \quad \lambda \in \mathbb{R}.$$

The plane Π_2 contains the point O and the line L .

- a. Given that L meets Π_1 at the point P, find the coordinates of P. [4]
- b. Find the shortest distance from the point O(0, 0, 0) to Π_1 . [4]
- c. Find the equation of Π_2 , giving your answer in the form $\mathbf{r} \cdot \mathbf{n} = d$. [3]
- d. Determine the acute angle between Π_1 and Π_2 . [5]



This question is based on AHL 3.18

Content	Guidance, clarification and syllabus links
Intersections of: a line with a plane; two planes; three planes.	Finding intersections by solving equations; geometrical interpretation of solutions.
Angle between: a line and a plane; two planes.	Link to: solutions of systems of linear equations (AHL 1.16).

- a. Given that L meets Π_1 at the point P , find the coordinates of P .

[4]

Step 1: Write the equation of Line in Parametric Form:

We have line L

$$\vec{r} = \begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix} + \lambda \begin{pmatrix} -3 \\ -1 \\ 4 \end{pmatrix}, \quad \lambda \in \mathbb{R}.$$

$$\begin{aligned} \therefore \quad x &= 1 - 3\lambda \\ y &= 2 - \lambda \\ z &= -2 + 4\lambda \end{aligned}$$

$$\begin{aligned} \vec{r} &= \vec{a} + \lambda \vec{b} \\ \vec{r} &= \begin{pmatrix} x \\ y \\ z \end{pmatrix} \end{aligned}$$



Step 2: Substitute into the Equation of Plane :

$$x = 1 - 3\lambda$$

$$y = 2 - \lambda$$

$$z = -2 + 4\lambda$$

The plane Π_1 has equation $3x - y + z = -13$

In equation of plane substituting values of x, y, z from equation of line, we get

$$3(1 - 3\lambda) - (2 - \lambda) + (-2 + 4\lambda) = -13$$

Step 3: Solve for the Parameter:

$$-9\lambda + \lambda + 4\lambda + 3 - 2 - 2 = -13$$

$$-4\lambda - 1 = -13$$

$$-4\lambda = -12$$

$$\lambda = 3$$



Step 4: Find the Coordinates of point of intersection :

substitute $\lambda = 3$, in the parametric equations of line

$$x = 1 - 3\lambda = 1 - 3(3) = -8$$

$$y = 2 - \lambda = 2 - 3 = -1$$

$$z = -2 + 4\lambda = -2 + 4(3) = 10$$

The coordinates of point of intersection P are $(-8, -1, 10)$

End of Lesson

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Note: Part b, c, and d we will solve in 2 more lessons.

IB AAHL

HOW TO FIND EQUATION OF A PLANE

IN VECTOR FORM

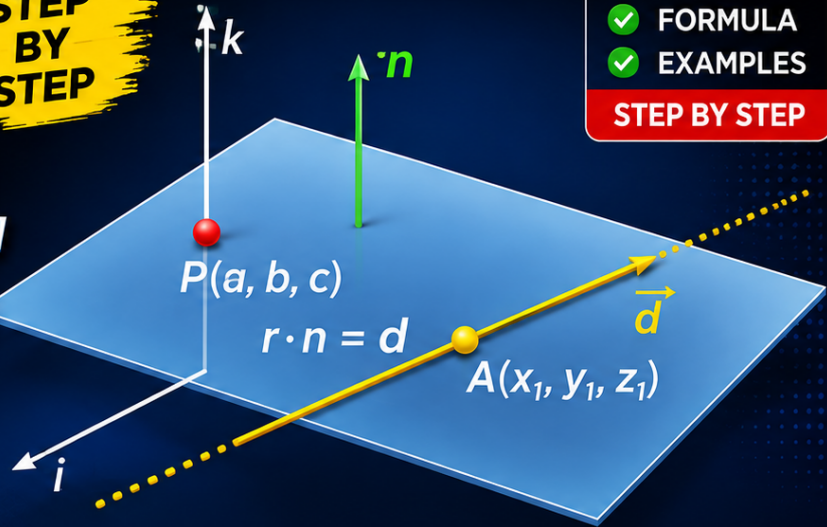
WHICH CONTAINS
A POINT AND A LINE

4 EASY STEPS

STEP
BY
STEP

- ✓ CONCEPTS
- ✓ FORMULA
- ✓ EXAMPLES

STEP BY STEP



1

TAKE A
POINT ON
THE LINE

2

USE DIRECTION
VECTOR OF
THE LINE

3

FIND NORMAL
VECTOR TO
THE PLANE

4

WRITE
EQUATION IN
VECTOR FORM





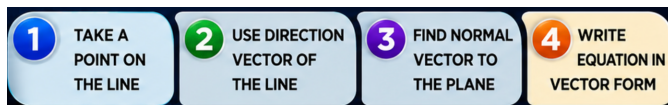
Question

The plane Π_1 has equation $3x - y + z = -13$ and the line L has vector equation

$$\mathbf{r} = \begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix} + \lambda \begin{pmatrix} -3 \\ -1 \\ 4 \end{pmatrix}, \quad \lambda \in \mathbb{R}.$$

The plane Π_2 contains the point O and the line L .

C. Find the equation of π_2 , giving your answer in the form $\vec{r} \cdot \vec{n} = d$

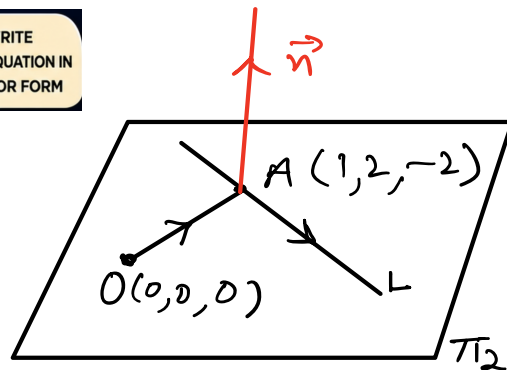


Step 1 : Identify a point in plane π_2 :

$A(1, 2, -2)$ is in π_2

also O is in π_2

$$\therefore \overrightarrow{OA} = \begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix}$$





Step 2 : Direction vector of line L in π_2 :

$$\vec{d} = \begin{pmatrix} -3 \\ -1 \\ 4 \end{pmatrix} \quad \left\{ r = \begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix} + \lambda \begin{pmatrix} -3 \\ -1 \\ 4 \end{pmatrix}, \lambda \in \mathbb{R}. \right\}$$

Step 3 : Normal vector to plane π_2 will be:

$$\vec{n} = \vec{OA} \times \vec{d}$$
$$\vec{n} = \begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix} \times \begin{pmatrix} -3 \\ -1 \\ 4 \end{pmatrix}$$

$$\therefore \vec{n} = \begin{pmatrix} 6 \\ 2 \\ 5 \end{pmatrix}$$

$$\vec{n} = \begin{vmatrix} i & j & k \\ 1 & 2 & -2 \\ -3 & -1 & 4 \end{vmatrix}$$

$$= (8 - 2)i - (4 - 6)j + (-1 + 6)k$$

$$\therefore \vec{n} = 6i + 2j + 5k$$



Step 4 : Equation of plane π_2 will be:

$$\vec{r} \cdot \vec{n} = d$$

$$\vec{r} \cdot \begin{pmatrix} 6 \\ 2 \\ 5 \end{pmatrix} = 0$$

If asked cartesian form

$$\left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \cdot \begin{pmatrix} 6 \\ 2 \\ 5 \end{pmatrix} = 0 \right\}$$
$$6x + 2y + 5z = 0$$

End of Lesson

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Question

The plane Π_1 has equation $3x - y + z = -13$ and the line L has vector equation

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-

Step 1: Identify the Components:

we have $O(0, 0, 0)$

→ & normal vector to plane $3x - y + z = -13$

$$\vec{n} = \begin{pmatrix} 3 \\ -1 \\ 1 \end{pmatrix}$$



Step 2: Find equation of line perpendicular to the plane from given point O:

$$\vec{r} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 3 \\ -1 \\ 1 \end{pmatrix} = \mu \begin{pmatrix} 3 \\ -1 \\ 1 \end{pmatrix}$$
$$\vec{r} = \mu \begin{pmatrix} 3 \\ -1 \\ 1 \end{pmatrix} \longrightarrow \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 3\mu \\ -\mu \\ \mu \end{pmatrix}$$

Step 3: Substituting into the plane the x, y & z components of this line:

$$3x - y + z = -13$$

$$3(3\mu) - (-\mu) + (\mu) = -13$$

$$\mu = -\frac{13}{11}$$



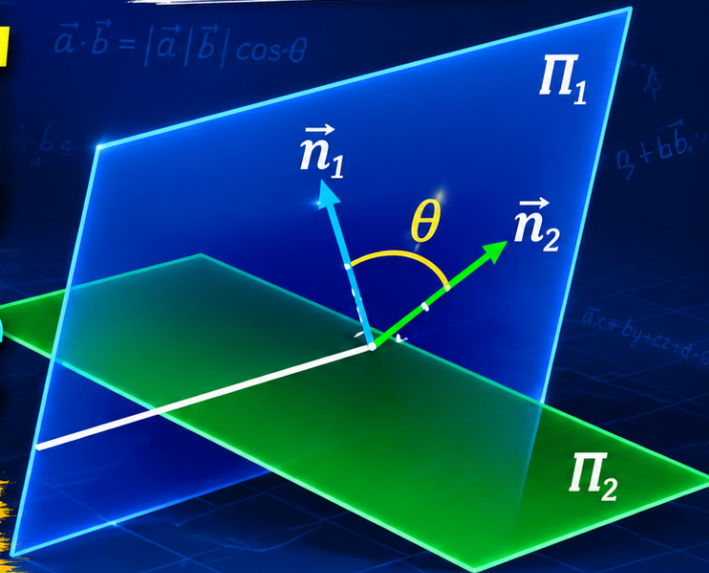
Step 4: Get the magnitude(length) of vector $\vec{r}$:

$$\begin{aligned}\mu &= -\frac{13}{11} \quad \longrightarrow \quad \vec{r} = \mu \begin{pmatrix} 3 \\ -1 \\ 1 \end{pmatrix} \\ \therefore \vec{r} &= \frac{-3}{11} \begin{pmatrix} 3 \\ -1 \\ 1 \end{pmatrix} = \begin{pmatrix} \frac{-9}{11} \\ \frac{3}{11} \\ \frac{-3}{11} \end{pmatrix} \\ \therefore |\vec{r}| &= \sqrt{\left(\frac{-9}{11}\right)^2 + \left(\frac{3}{11}\right)^2 + \left(\frac{-3}{11}\right)^2} \\ &= \frac{\sqrt{81 + 9 + 9}}{11} \\ &= \frac{\sqrt{99}}{11} \\ &= \frac{3\sqrt{11}}{11}\end{aligned}$$

HOW TO FIND ANGLE BETWEEN TWO PLANES

IN VECTOR FORM

$$\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \theta$$



$$\cos \theta = \frac{|\vec{n}_1 \cdot \vec{n}_2|}{|\vec{n}_1| |\vec{n}_2|}$$

IN **4**
EASY STEPS



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Question

The plane Π_1 has equation $3x - y + z = -13$ and the line L has vector equation

$$r = \begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix} + \lambda \begin{pmatrix} -3 \\ -1 \\ 4 \end{pmatrix}, \quad \lambda \in \mathbb{R}.$$

The plane Π_2 contains the point O and the line L .

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d. Determine the acute angle between Π_1 and Π_2 .

[5]

Step 1: Extract the normal vectors :

$$3x - y + z = -13$$

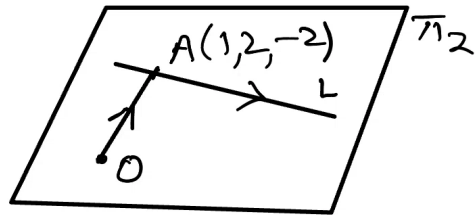
is

$$\vec{n}_1 = \begin{pmatrix} 3 \\ -1 \\ 1 \end{pmatrix}$$



Plane π_2 contains vector $\overrightarrow{OA} = \begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix}$

and direction vector L is $\vec{d} = \begin{pmatrix} -3 \\ -1 \\ 4 \end{pmatrix}$



$$\therefore \vec{n}_2 = \overrightarrow{OA} \times \vec{d}$$

$$\therefore \vec{n}_2 = \begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix} \times \begin{pmatrix} -3 \\ -1 \\ 4 \end{pmatrix} = \begin{pmatrix} 6 \\ 2 \\ 5 \end{pmatrix}$$

Vector product

$$\mathbf{v} \times \mathbf{w} = \begin{pmatrix} v_2 w_3 - v_3 w_2 \\ v_3 w_1 - v_1 w_3 \\ v_1 w_2 - v_2 w_1 \end{pmatrix}, \text{ where } \mathbf{v} = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix}, \mathbf{w} = \begin{pmatrix} w_1 \\ w_2 \\ w_3 \end{pmatrix}$$

$$= \begin{pmatrix} (2)(4) - (-2)(-1) \\ (-2)(-3) - (1)(4) \\ (1)(-1) - (2)(-3) \end{pmatrix}$$



Step (2): Calculate the dot product :

$$\vec{n}_1 = \begin{pmatrix} 3 \\ -1 \\ 1 \end{pmatrix} \quad \text{and} \quad \vec{n}_2 = \begin{pmatrix} 6 \\ 2 \\ 5 \end{pmatrix}$$

$$\begin{aligned} \therefore \vec{n}_1 \cdot \vec{n}_2 &= \begin{pmatrix} 3 \\ -1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 6 \\ 2 \\ 5 \end{pmatrix} = (3)(6) + (-1)(2) + (1)(5) \\ &= 18 - 2 + 5 = 21 \end{aligned}$$



Step (3) : Find the magnitudes:

$$|\vec{n}_1| = \left| \begin{pmatrix} 3 \\ -1 \\ 1 \end{pmatrix} \right| = \sqrt{(3)^2 + (-1)^2 + (1)^2} = \sqrt{11}$$

$$|\vec{n}_2| = \left| \begin{pmatrix} 6 \\ 2 \\ 5 \end{pmatrix} \right| = \sqrt{(6)^2 + (2)^2 + (5)^2} = \sqrt{65}$$



Step (4) : Solve for the angle:

Let θ be the angle between π_1 and π_2

$$\begin{aligned}\cos \theta &= \frac{\vec{n}_1 \cdot \vec{n}_2}{|\vec{n}_1| |\vec{n}_2|} \\ &= \frac{21}{\sqrt{11} \sqrt{65}} \\ \theta &= \cos^{-1} \left(\frac{21}{\sqrt{11} \sqrt{65}} \right) \approx 0.668\end{aligned}$$

End of Lesson

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You can download the pdf of these 3. Lessons from the following link:

<https://github.com/nasirmaths/IB-YouTube-pdf/blob/main/>